University of Illinois **Spring 2025**

ECE 313: Problem Set 2: Problems and Solutions

Due: Friday, February 14 at 07:00:00 p.m.
Reading: *ECE 313 Course Notes,* Sections 2.3, 2.10, 2.4.1 - 2.4.2

Note on reading: For most sections of the course notes there are short answer questions at the end of the chapter. We recommend that after reading each section you try answering the short answer questions. Do not hand these in; answers to the short answer questions are provided in the appendix of the notes.

Note on turning in homework: Homework is assigned on a weekly basis on Fridays, and is due by 7 p.m. on the following Friday. You must upload handwritten homework to Gradescope. Alternatively, you can typeset the homework in LaTeX. However, no additional credit will be awarded to typeset submissions. No late homework will be accepted.

Please write on the top right corner of the first page:
NAME
NETID
SECTION
PROBLEM SET #

Page numbers are encouraged but not required. Five points will be deducted for improper headings. Please assign your uploaded pages to their respective question numbers while submitting your homework on Gradescope. **5 points will be deducted for incorrectly assigned pages.**

1. **[Choosing Shirts]**
   A bag contains 6 orange and 3 blue t-shirts. Two t-shirts are randomly taken out of that bag and moved into a second bag, which already contains an orange t-shirt. Then a t-shirt is chosen from the second bag at random.

   (a) Let X denote the number of blue t-shirts moved from the first bag into the second bag. Find the pmf of X.
   Solution: X can be either 0, 1, or 2. There are $\binom{9}{2}$ ways to choose 2 shirts out of the 9 in the bag.

$$P(X=0) = \frac{\binom{6}{2}}{\binom{9}{2}} = \frac{5}{12}$$

$$P(X=1) = \frac{\binom{6}{1}\binom{3}{1}}{\binom{9}{2}} = \frac{1}{2}$$

$$P(X=2) = \frac{\binom{3}{2}}{\binom{9}{2}} = \frac{1}{12}$$

   (b) What is the probability the t-shirt chosen from the second bag is blue?
   Solution: No matter what, there will be three t-shirts in the second bag. Let A denote the event of getting a blue t-shirt from the second bag. By the law of total probability,

$$
\begin{aligned}
P(A) &= P(A|X=0)P(X=0) + P(A|X=1)P(X=1) + P(A|X=2)P(X=2) \\
&= 0 \cdot \frac{5}{12} + \frac{1}{3} \cdot \frac{1}{2} + \frac{2}{3} \cdot \frac{1}{12} \\
&= \frac{2}{9}
\end{aligned}
$$

(c) What is the conditional probability of $X = 1$, given the t-shirt drawn from the second bag is blue?

Solution: By Bayes' formula,

$$
\begin{aligned}
P(X=1|A) &= \frac{P(A|X=1)P(X=1)}{P(A)} \\
&= \frac{\left(\frac{1}{3} \cdot \frac{1}{2}\right)}{\frac{2}{9}} \\
&= \frac{3}{4}
\end{aligned}
$$

2. **[Rolling Dice I]**

Suppose you have three fair dice D_1, D_2, and D_3. D_1 has two faces painted green, D_2 has three faces painted green, and D_3 has five faces painted green. You randomly choose one of these dice and roll it repeatedly.

(a) Find the probability that the first roll shows a green face.

Solution: Let A denote the event that the first roll shows a green face. By the law of total probability,

$$
\begin{aligned}
P(A) &= P(A|D_1)P(D_1) + P(A|D_2)P(D_2) + P(A|D_3)P(D_3) \\
&= \left(\frac{2}{6} + \frac{3}{6} + \frac{5}{6}\right) \times \frac{1}{3} \\
&= \frac{5}{9}
\end{aligned}
$$

(b) Given that the first roll showed a green face, find the conditional probability that the second roll also shows a green face.

Solution: Let B denote the event that the second roll shows a green face. Rolls of a given die are independent, so $P(AB|D) = P(A|D)P(B|D)$. With that in mind,

$$
\begin{aligned}
P(B|A) &= \frac{P(AB)}{P(A)} \\
&= \frac{P(AB|D_1)P(D_1) + P(AB|D_2)P(D_2) + P(AB|D_3)P(D_3)}{P(A)} \\
&= \frac{P(A|D_1)P(B|D_1)P(D_1) + P(A|D_2)P(B|D_2)P(D_2) + P(A|D_3)P(B|D_3)P(D_3)}{P(A)} \\
&= \frac{\left(\frac{2}{6} \cdot \frac{2}{6} + \frac{3}{6} \cdot \frac{3}{6} + \frac{5}{6} \cdot \frac{5}{6}\right) \times \frac{1}{3}}{\frac{5}{9}} \\
&= \frac{19}{30}
\end{aligned}
$$

Note that we CANNOT say that $P(AB) = P(A)P(B)$. For intuition, if A occurs, it would make you lean towards the die being D_3, which is additional information for whether or not B occurs. Only when you control for the die are the events independent.

(c) Given that the first roll showed a green face, find the conditional probability that the second roll doesn't show a green face.

Solution: Note that we have to find $P(B^c|A)$, so

$$P(B^c|A) = 1 - P(B|A) = \frac{11}{30}$$

You may double-check using a similar approach as in (b).

3. **[Rolling Dice II]**

A fair die is rolled repeatedly. Each time the die is rolled, the number is written down. Let X be the number of rolls until some number shows twice, thus $X \in \{2, 3, \ldots, 7\}$.

(a) Find $P(X = 4)$.

Solution: $X = 4$ means that the first 3 rolls show distinct numbers ($6 \times 5 \times 4$ ways), and the number of the fourth roll must have appeared in the previous 3 rolls (3 ways). Therefore,

$$P(X = 4) = \frac{(6 \times 5 \times 4) \times 3}{6^4} = \frac{5}{18}$$

(b) Find $P(X = 6|X > 3)$.

Solution: $X > 3$ means that the first 3 rolls show distinct numbers, so

$$P(X > 3) = \frac{6 \times 5 \times 4}{6^3} = \frac{5}{9}$$

Then by the formula for conditional probability:

$$\begin{aligned} P(X = 6|X > 3) &= \frac{P(X = 6, X > 3)}{P(X > 3)} \\ &= \frac{P(X = 6)}{P(X > 3)} \\ &= \frac{(6 \times 5 \times 4 \times 3 \times 2) \times 5}{6^6} \times \frac{9}{5} \\ &= \frac{5}{36} \end{aligned}$$

Alternatively, we can assume we have already rolled 3 times and got distinct numbers. Then the fourth and fifth rolls must be two distinct numbers not appeared in the previous 3 rolls (3×2 ways), and the number of the sixth roll must have appeared in the previous 5 rolls (5 ways). This yields $P(X = 6|X > 3) = \frac{(3\times2)\times5}{6^3} = \frac{5}{36}$.

4. **[Independence]**

Suppose two fair dice, one orange and one blue, are rolled. Define the following events:

A: The product of the two numbers that show is 4.

B: The number on the orange die is strictly smaller than the number on the blue die.

C: The sum of the numbers is greater than or equal to the product.

D: The number on the blue die is 2.

(a) List all pairs of events, if any, from the set A, B, C, and D that are independent.

Solution: We write (i, j) for the outcome that the orange die shows i and the blue die shows j. For each event,

$$A = \{(1,4),(2,2),(4,1)\},$$
$$B = \{(i,j) : 1 \le i < j \le 6\}$$
$$C = \{(1,j) : 1 \le j \le 6\} \cup \{(i,1) : 1 \le i \le 6\} \cup \{(2,2)\},$$
$$D = \{(i,2) : 1 \le i \le 6\}.$$

Then we can compute the following probabilities:

$$P(A) = \frac{1}{12}, \quad P(B) = \frac{5}{12}, \quad P(C) = \frac{1}{3}, \quad P(D) = \frac{1}{6}$$
$$P(AB) = P\{(1,4)\} = \frac{1}{36}, \ P(AC) = P(A) = \frac{1}{12},$$
$$P(AD) = P\{(2,2)\} = \frac{1}{36}, \ P(BC) = P\{(1,j) : 2 \le j \le 6\} = \frac{5}{36},$$
$$P(BD) = P\{(1,2)\} = \frac{1}{36}, \ P(CD) = P\{(1,2),(2,2)\} = \frac{1}{18}$$

So the pairs (B, C) and (C, D) are independent; no other pair is independent.

(b) List all triplets of events, if any, from the set of A, B, C, and D that are mutually independent.

Solution: No three events are even pairwise independent by part (a), so no three events are mutually independent.

5. **[Flipping a Coin]**
Mary and Joe are alternately flipping a fair coin, starting with Mary.

(a) Let X denote the total number of heads showing in the first 2 flips. Assume the coin flips are independent. Find $p_X(u)$, the pmf of X.

Solution: The sample space can be listed as {HH, HT, TH, TT}, where each outcome is equally likely to occur.

$$p_X(0) = \frac{1}{4}, \ p_X(1) = \frac{1}{2}, \ p_X(2) = \frac{1}{4}$$

(b) Now suppose that Joe decides to impress Mary by using a trick coin for their game. This coin has a unique property: it somehow matches the result of Mary's most recent flip with probability 0.8. Mary is still using a fair coin. Let Y denote the total number of heads showing in the first 4 flips, with Mary using a fair coin and Joe using his trick coin. Find $p_Y(v)$, the pmf of Y.

Solution: Since Joe's flips are dependent on Mary's while Mary's flips are not affected by Joe's, we may consider two consecutive flips (first Mary, then Joe) together as a unit, while different units are independent since they start with Mary's turn. Let H_M and T_M denote Mary's result, while H_J and T_J denote Joe's result.

We have the following probabilities for each outcome for a unit:

$$
\begin{aligned}
P(\mathrm{HH}) &= P(H_J|H_M)P(H_M) = \frac{4}{5}\cdot\frac{1}{2} = \frac{2}{5} \\
P(\mathrm{HT}) &= P(T_J|H_M)P(H_M) = \frac{1}{5}\cdot\frac{1}{2} = \frac{1}{10} \\
P(\mathrm{TH}) &= P(H_J|T_M)P(T_M) = \frac{1}{5}\cdot\frac{1}{2} = \frac{1}{10} \\
P(\mathrm{TT}) &= P(T_J|T_M)P(T_M) = \frac{4}{5}\cdot\frac{1}{2} = \frac{2}{5}
\end{aligned}
$$

Then the first 4 flips correspond to the first 2 units, and thus

$$
\begin{aligned}
p_Y(0) &= P(\mathrm{TTTT}) = P(\mathrm{TT})P(\mathrm{TT}) = \frac{4}{25} \\
p_Y(1) &= P(\mathrm{HTTT}) + P(\mathrm{THTT}) + P(\mathrm{TTHT}) + P(\mathrm{TTTH}) \\
&= 4\cdot\frac{2}{5}\cdot\frac{1}{10} = \frac{4}{25} \\
p_Y(2) &= P(\mathrm{HHTT}) + P(\mathrm{THTH}) + P(\mathrm{THHT}) + P(\mathrm{HTTH}) + P(\mathrm{HTHT}) + P(\mathrm{TTHH}) \\
&= 2\cdot\frac{2}{5}\cdot\frac{2}{5} + 4\cdot\frac{1}{10}\cdot\frac{1}{10} = \frac{9}{25} \\
p_Y(3) &= P(\mathrm{HHHT}) + P(\mathrm{HHTH}) + P(\mathrm{HTHH}) + P(\mathrm{THHH}) \\
&= 4\cdot\frac{2}{5}\cdot\frac{1}{10} = \frac{4}{25} \\
p_Y(4) &= P(\mathrm{HHHH}) = \frac{4}{25}
\end{aligned}
$$